

SENTINEL v2.0.0

Line of Defense Engine -- Overview Cheatsheet

TrinTech Digital Defense | github.com/trintechdigitaldefense/Sentinel

Repo	github.com/trintechdigitaldefense/Sentinel
Version	2.0.0
Lines	1,514 (zero external dependencies)
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CORE COMMANDS

Command	Description
start [-v]	Full scan + deception + Line of Defense
scan [-v]	Quick network scan with nmap
report	Full report: devices, alerts, defense status
status	Current Line of Defense module status
config [-v]	Initialize or view config
alerts	Recent security alerts
deception	Deploy deception artifacts
hunt [-v]	Persistence hunting loop (Ctrl+C to stop)
info	System info and defense module status
clean	Reset all tracking data

LINE OF DEFENSE COMMANDS

Command	Description
integrity	Generate file integrity baseline (SHA-256)
integrity --check	Check integrity against existing baseline
watch [-v]	Real-time filesystem watcher (inotify)
processes	Scan for suspicious processes
outbound	Monitor outbound connections
blocklist	Show blocked IPs
unblock	Unblock an IP, remove iptables rule
users	Show user accounts and sudo
cron	Check cron integrity
hardening	System hardening audit (0-100 score)

DEFENSE MODULES

Module	Detects	Alert
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FIM -- Filesystem Integrity	File mods/deletions/new files, checksum changes	FIM_MODIFIED, FIM_DELETED, FIM_REALTIME
Process Monitor	Cryptominers, malware paths, high CPU >50%	SUSPICIOUS_PROCESS, MALWARE_PATH_PROCESS, HIGH_CPU_PROCESS
SSH Brute Force + Auto-Block	Failed auth, auto iptables blocking (5 fail, 30min TTL)	IP_BLOCKED, AUTH_FAILURE
Outbound Monitor	Reverse shell ports (4444, 5555, 31337), foreign IPs	OUTBOUND_SUSPICIOUS_PORT
User Monitor	Sudo usage, failed sudo, new users	SUDO_USAGE, FAILED_SUDO
Cron Monitor	New/removed crontab entries	NEW_CRON, REMOVED_CRON
Hardening Audit	SSH config, firewall, world-writable files, SUID, 0-100 score, regression	SECURITY_REGRESSION

HARDENING AUDIT -- 0-100 SCORE

- bullet SSH PermitRootLogin yes: critical (-10pts)
- bullet SSH PasswordAuthentication yes: warning (-5pts)
- bullet No iptables rules: critical (-15pts)
- bullet World-writable config files >3: warning (-3pts)
- bullet SUID binaries >20: warning (-5pts)
- bullet Core dumps to /tmp/: warning (-3pts)
- bullet Empty passwords: critical (-10pts)
- bullet Regression: score drop >10: alert

SYSTEM REQUIREMENTS

- bullet Python 3.6+ -- stdlib only (no pip installs)
- bullet nmap -- network scanning
- bullet inotify-tools -- real-time file watcher
- bullet iptables -- SSH brute force blocking
- bullet journalctl or /var/log/auth.log -- SSH monitoring
- bullet Read access to /etc/passwd, /etc/shadow, /proc/net/tcp

CONFIGURATION

File: `~/sentinel/config.json`

All modules toggled via config. Each has **active** flag + module settings.

TYPICAL WORKFLOW

1. **Setup** -- *sentinel start*: init config, scan network, deploy deception
2. **Baselines** -- *sentinel integrity* + *sentinel cron*: create baselines
3. **Monitor** -- *sentinel hunt*: continuous background monitoring
4. **On-Demand** -- *processes*, *outbound*, *hardening*, *users*, *cron*
5. **Review** -- *sentinel alerts*: recent security events
6. **Report** -- *sentinel report*: full defense status

ALERT TYPES

Level	Event	Description
Network	NEW_DEVICE	Unauthorized device on network
Deception	DECOY_MODIFIED	Honeypot artifact modified

Integrity	FIM_MODIFIED	Critical file checksum changed
Integrity	FIM_DELETED	Critical file deleted
Integrity	FIM_REALTIME	Real-time file change
Process	SUSPICIOUS_PROCESS	Suspicious process detected
Process	MALWARE_PATH_PROCESSES	Process from malware path
Process	HIGH_CPU_PROCESS	Process >50% CPU
Network	AUTH_FAILURE	SSH auth failures
Network	IP_BLOCKED	IP auto-blocked via iptables
Network	OUTBOUND_SUSPICIOUS_PORT	Suspicious outbound port
User	SUDO_USAGE	Sudo command executed
User	FAILED_SUDO	Failed sudo attempt
Cron	NEW_CRON	New cron entry
Cron	REMOVED_CRON	Cron entry removed
System	SECURITY_REGRESSION	Hardening score dropped >10
System	LOG_TAMPERING	Log file deleted/modified

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